



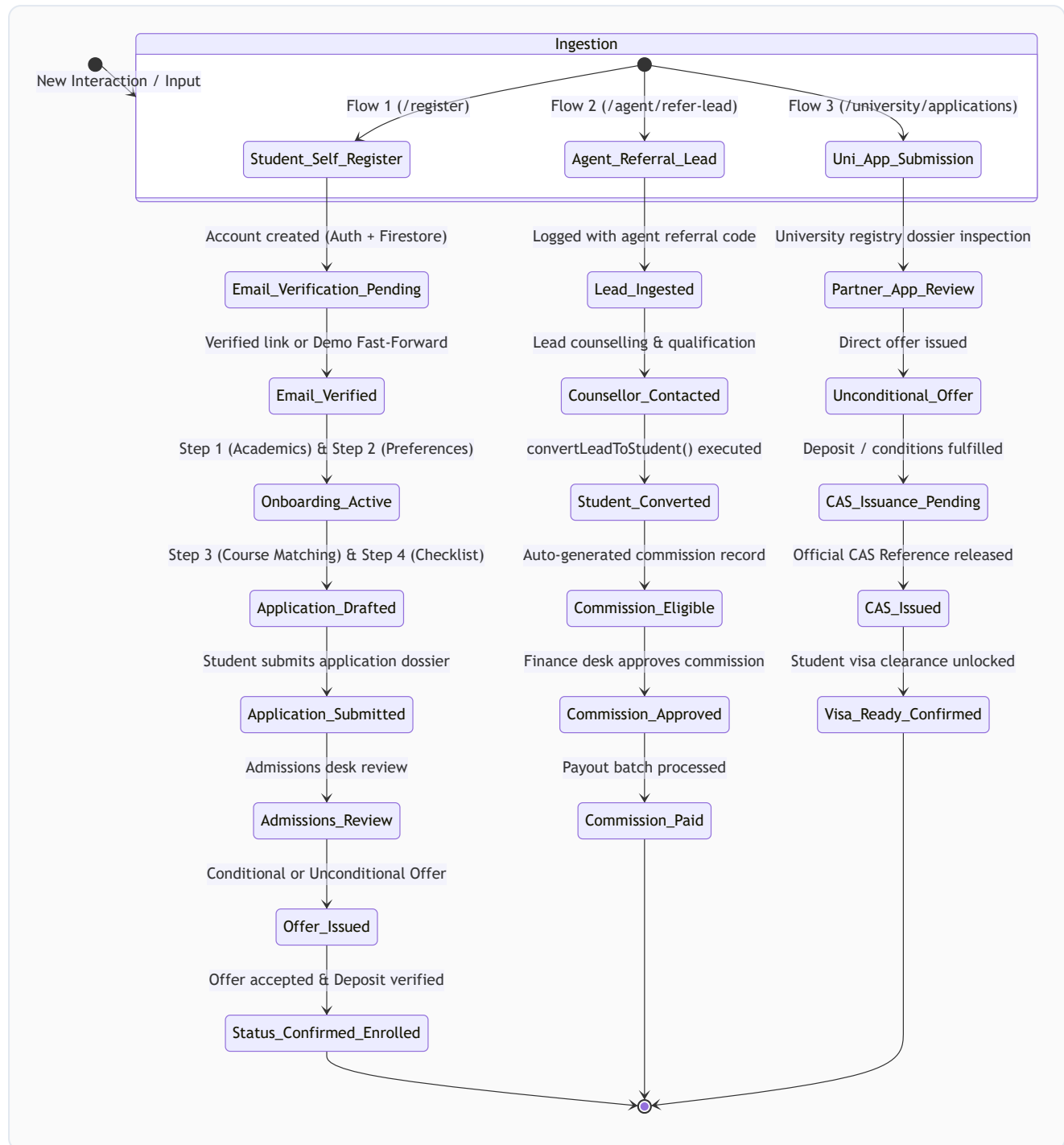
EduCRM System Workflow & Architecture Specification

Comprehensive Documentation of State Transitions, BPMN Multi-Role Workflows, System Sequence Diagrams (SSD), and Component Sequence Diagrams (SD).

Version 2.4.0 • Enterprise Architecture Reference

1. State Transition Diagram — Core Multi-Flow Lifecycles

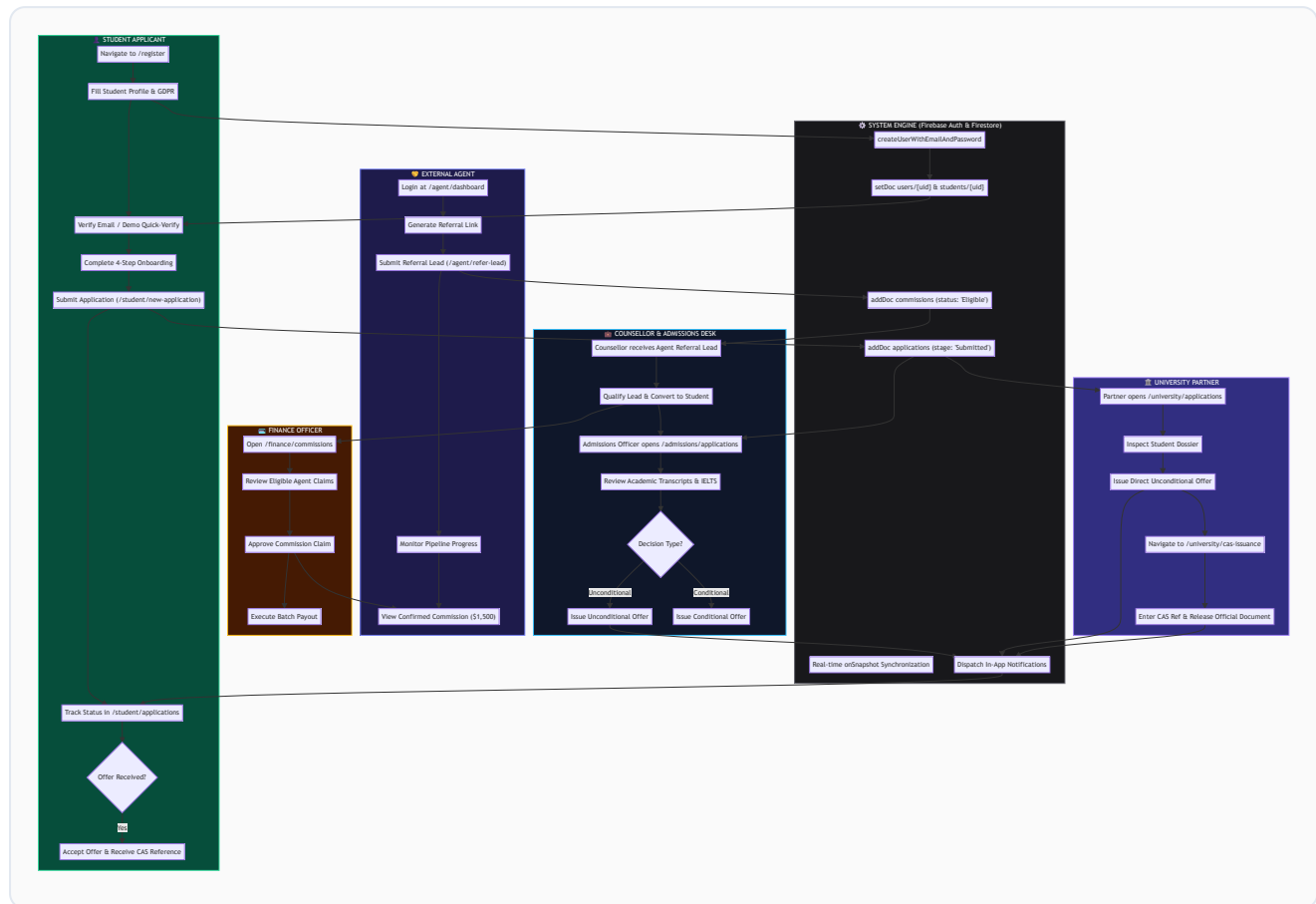
Documents all discrete operational states from initial ingestion through intermediate review, offer generation, compliance clearance, and confirmed final status across the 3 primary system workflows.



Core Transition Rules: Every state transition triggers both an updated document attribute in Cloud Firestore and an immutable audit event in `audit_logs`. In-app student and partner notifications are dispatched whenever an application enters `Unconditional Offer`, `Approved`, or `CAS Issued`.

2. BPMN Multi-Role Process Swimlane Diagram

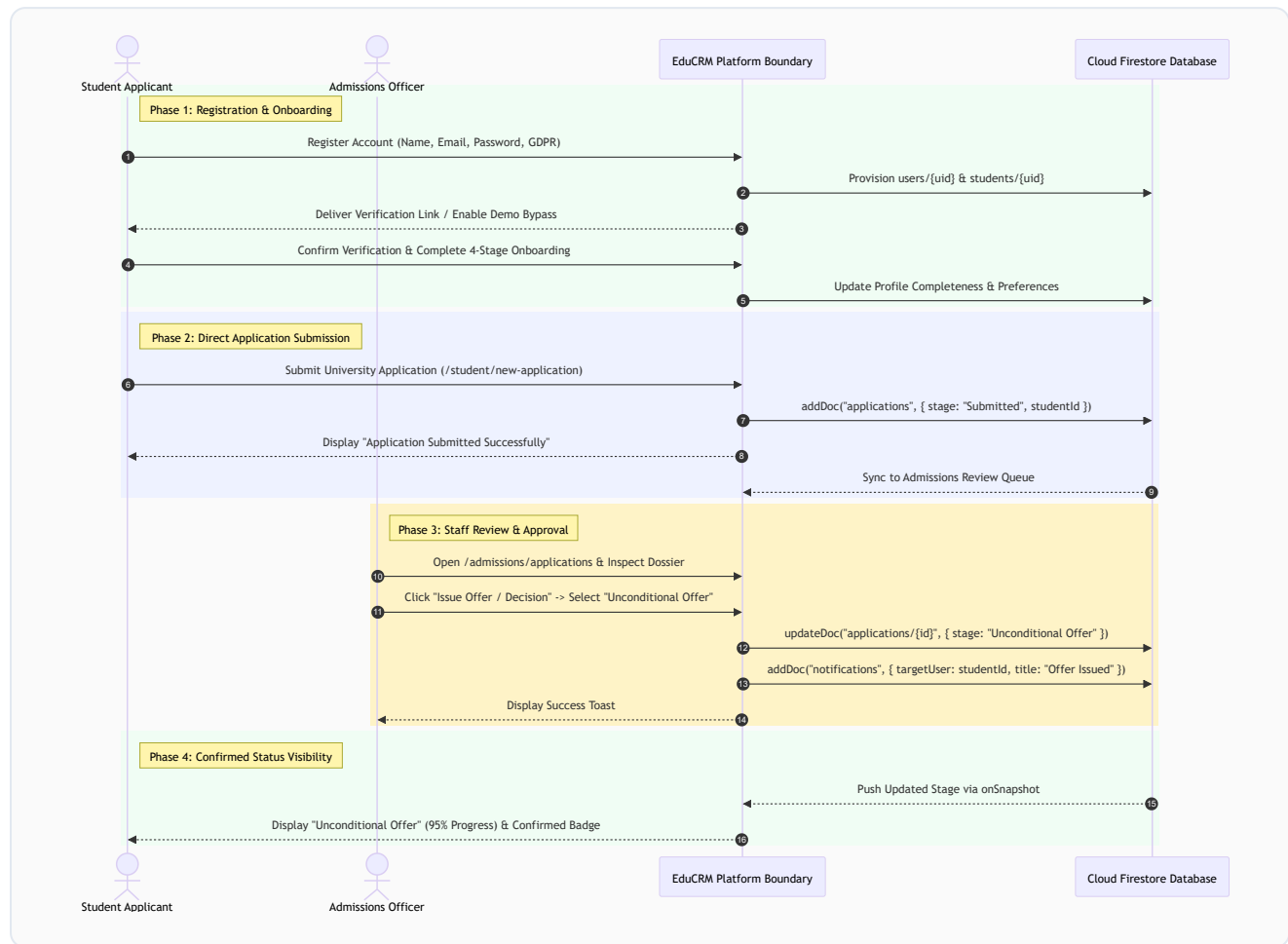
Business Process Model and Notation (BPMN) swimlane diagram illustrating cross-functional handoffs between Student, External Agent, Counsellor, Admissions Officer, University Partner, Finance Officer, and the System Backend.



Enterprise Coordination: Real-time Firestore synchronization guarantees that actions taken by the Admissions Officer or University Partner immediately reflect in the Student Portal and Agent Ledger without manual page refreshes.

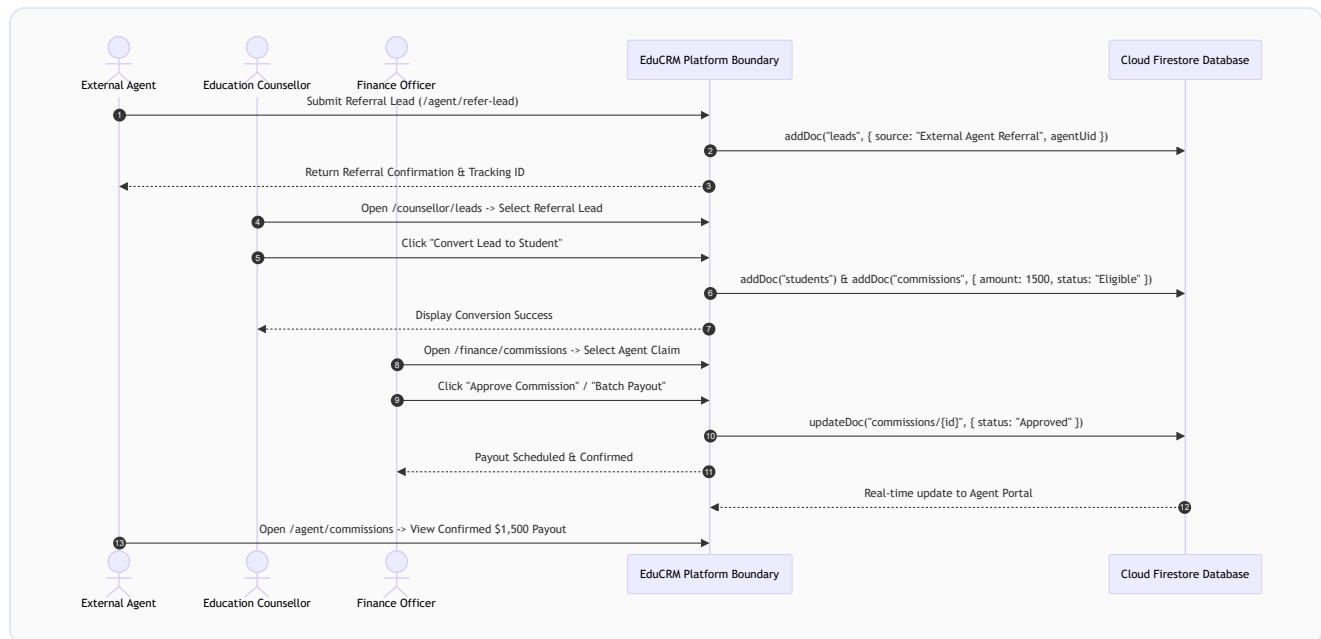
3.1 System Sequence Diagram (SSD) — Student Registration to Approval

High-level interaction between the Student Applicant, Admissions Officer, and the EduCRM Platform Boundary.



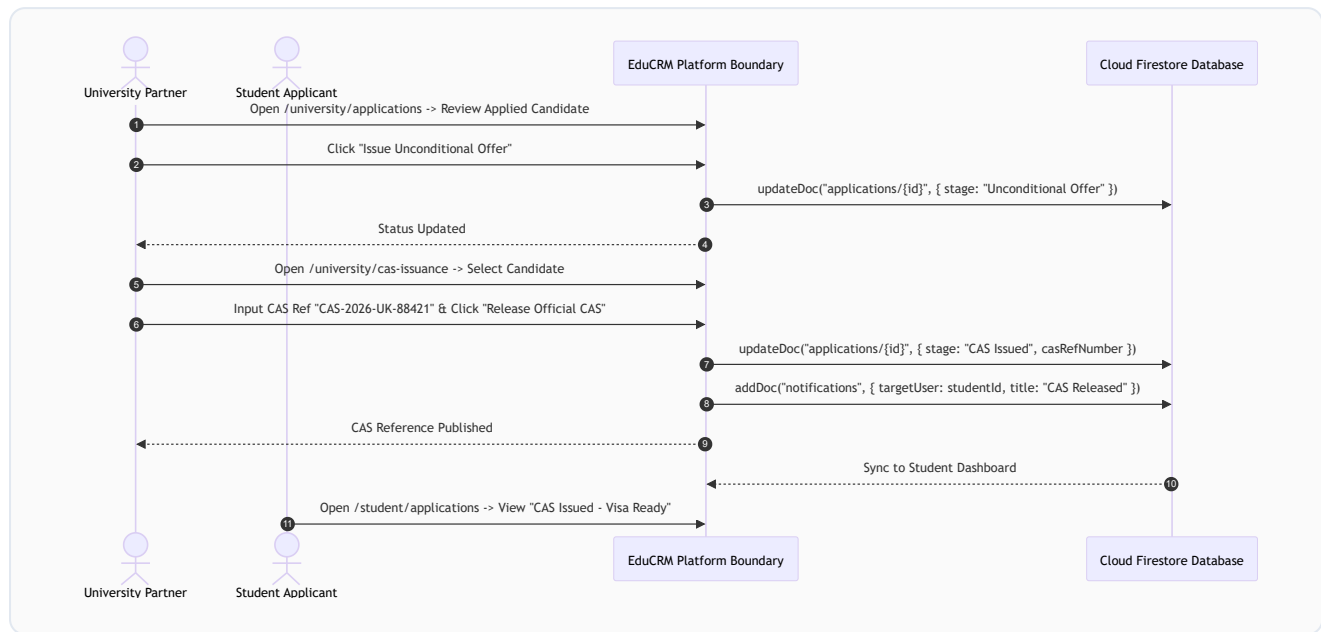
3.2 System Sequence Diagram (SSD) — Agent Referral to Commission Confirmed

High-level interaction between the External Agent, Counsellor, Finance Officer, and EduCRM.



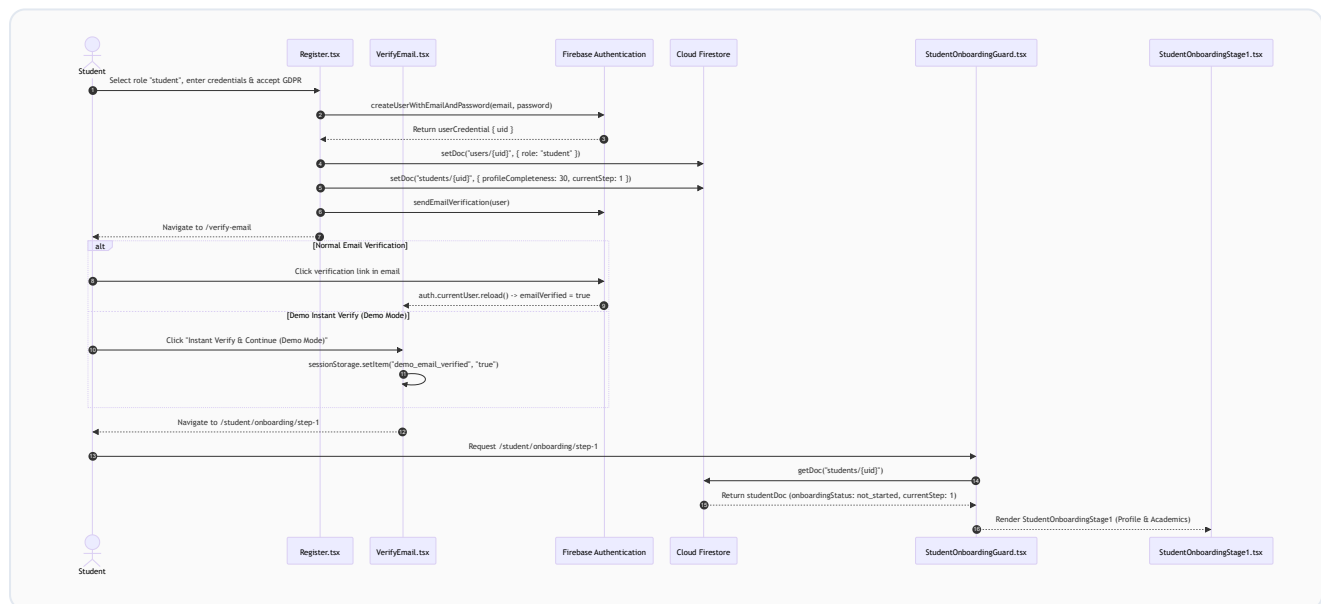
3.3 System Sequence Diagram (SSD) — University Partner Offer & CAS Release

Interaction between University Partner, Registry Officer, and Student for direct CAS release.



4.1 Component Sequence Diagram (SD) — Student Registration & Verification

Component-level trace showing React UI, Firebase Auth, Firestore, and StudentOnboardingGuard.



4.2 Component Sequence Diagram (SD) — Admissions Decision & Real-Time Sync

Detailed execution trace of AdmissionsWorkspace, useAdmissionsData, GlobalDataContext, and StudentApplications.

